QC Story Project

Project Title:

To eliminate the Wheel Speed Sensor (WSS) fail issue in Roller test for Scorpio model.

Automotive Division, Nashik Plant - 1 , STCF CAI Dept

	Plan	Actual
Start Date	02.03.19	02.03.19
Completion Date	25.10.19	25.10.19

Current Status:

Team	Name	Dept
Leader	**Mahesh A Savkar** Email: 23075562@mahindra.com	**CAI**
Members	Prashant tarte	CDMM
	Santosh Jadhav	MQA
	Nirmal Sambre	SQA
	Satish Pagar	MQA
Dept Head	Rahul Narkar	HOD
Mentor	Srikant Mulay	

QC Story Steps

1. Select the Theme

1. Validate Selection
2. Define The problem
3. Form a team
4. Make A project Plan

2. Justify the Choice & Containment action

1. Past trend of the problem
2. Pareto Diagram
3. Impact of the problem
4. Cost of Poor Quality
5. Containment action

3. Understand Current Situation

1. Process Flow Diagram
2. Past data – Different View point
3. Gemba Observations
4. Quick win Identification
5. Results of Quick wins

4. Decide Target

5. Analyze

1. List the SSVs
2. Identify the tool
3. Collect the data
4. Test the Hypothesis
5. Conclusions of analysis – *Decide funneling needed or not*
6. Funneling to next level Cause

6. Implement Corrective Measures

1. Generate Possible Solutions
2. Evaluate and Select Solution
3. Implement the action

7. Confirm Results and effects

1. Collect data
2. Capture all effects
3. Compare before and after
4. Measure Benefits

8. Standardize

1. Update Documents
2. Train people to follow new standards
3. Audit standards followed or not

9. Summarize and Future Actions

1. Document Learning
2. Horizontal deployment
3. Future action

1. Select the theme

1.1 Validate Selection
1.2 Define The problem
1.3 Form a team
1.4 Make A project Plan

1.1 Validate Selection Risk Assessment

No.	Success predictors	Yes (2)	Partial (1)	No (0)
1	**Project is a strategic priority, aligned organization goal (Big Y)**	2		
2	**Identify project area of focus :** Be sure there is no overlap with other MYB/MGB/MBB projects	2		
3	**Scope of the project is under the area of my influence**	2		
4	**Project scope is limited to only one product / one type of defect**	2		
5	**It is related to repetitive problem (chronic, not sporadic)**	2		
6	**Root cause is unknown and solution is unknown** (Type 4 problem)	2		
6	**Current level of the project is measured**	2		
7	**Potential Benefits of the project is identified**	2		
8	**Right vertical & horizontal team members are available**	2		
9	**Team members include technical experts**	2		
10	**Management is willing to commit serious resources to solutions**	2		
11	**Project is capable of completion within 4~6 months**	2		
12	**Project registration form is approved by HOD**	2		
	*** Any factor less than "2" creates a risk**			

Total: 24

Success probability : High=22~24 Medium=19~21 Low=1~18

1.2 Define the problem

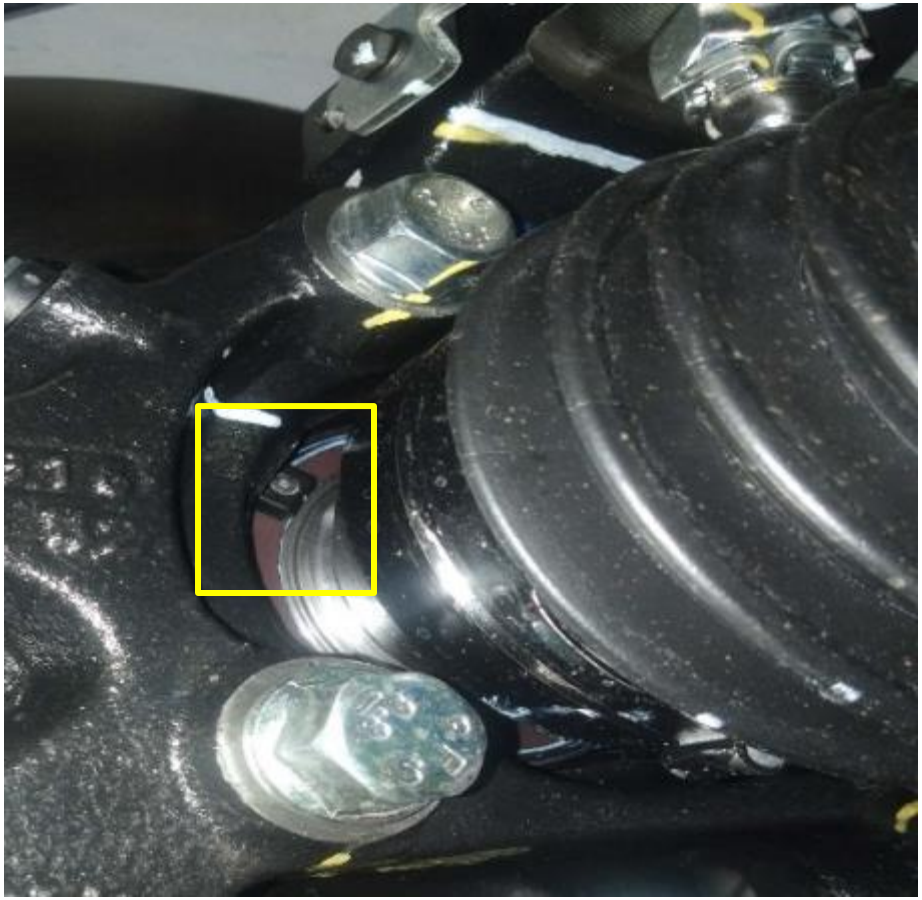
2. 5W 1H of the problem:

5W 1H	Details
What	WSS parameter fail in Roller test.
Where	At CAI roller test and rattle test stage.
Who	Reported by the Roller associates , rattle associates.
When	From Jan-2019
Why	V1+ Severity concern Leading to STR loss , FRC loss , RTR loss
How	Average 4-5 vehicles per day reported in CAI.

1.2 Define the problem

1. Brief Description of the problem:

Wheel speed sensor function



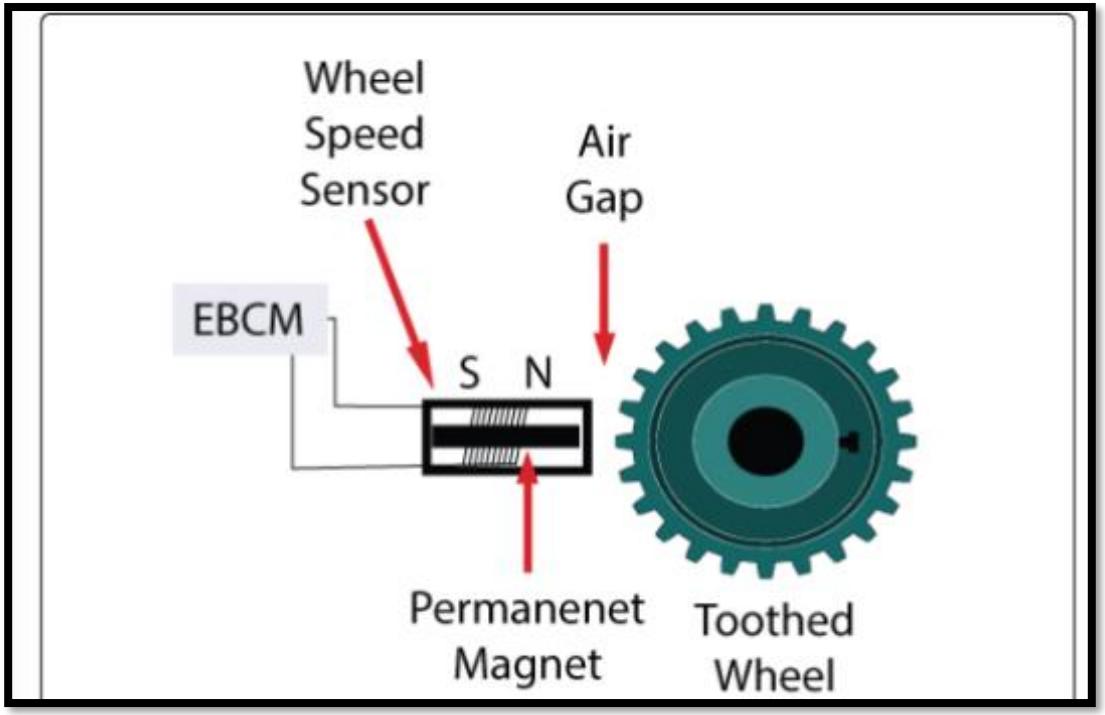
The anti-lock braking system is designed to apply smooth application of brake fluid to engage the brake calipers and pads to slow your car without locking up the tires. The wheel **speed sensor** is responsible for relaying the **wheel speed** to the ECU so that it can tell the **ABS** system how much pressure to safely apply.

1.2 Define the problem

1. Brief Description of the problem:

How wheel speed sensor works

The ABS controller uses wheel speed sensors to detect the rotational speed of each wheel. They're a permanent magnet sensor; a coil of wire wound around a metal core. The magnetic lines of flux created by this sensor change as a toothed ring rotates past its tip. They create voltage pulses that increase with wheel speed

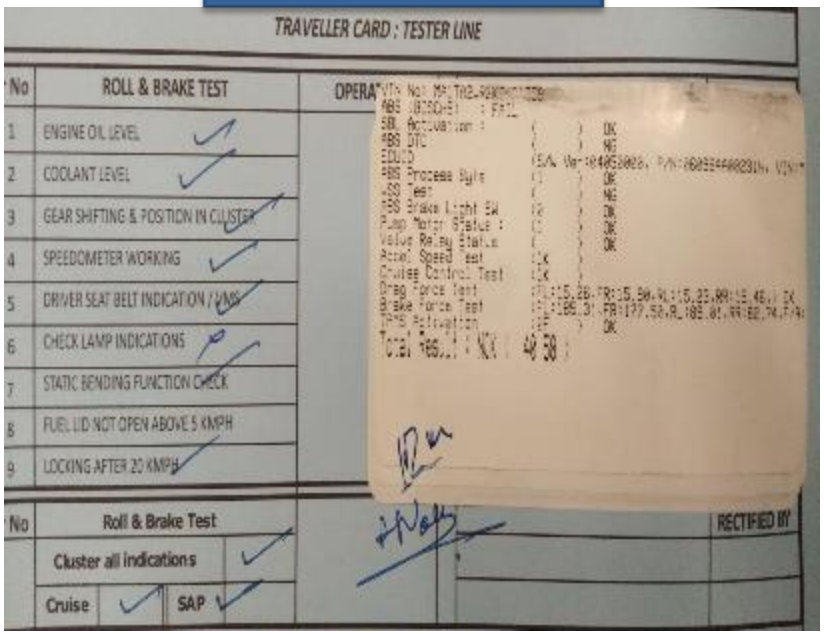


1.2 Define the problem

1. Brief Description of the problem:

During Roller test in RBT , vehicle runs in idle condition at roller . Samely the drum rotates and this signal is been taken by encoder, and it gives the output . which is been match in machine w.r.t WSS sensor signals. During roller test this wss parameter gets fail , and it gives roller fail result on sticker as NOK , and its error code as 39,40.

Traveller Card



Roller machine cycle test

3. Problem details

Sl	Check points	Details
1	Vehicle Model Selected for project	Scorpio All models
2	Assembly / Part selected for project	1. Wheel speed sensor. 2. LCA.
3	Other Similar Models having the problem	No
4	Other similar models not having the problem:	No issue in Marazzo, Xylo as The design in these models is different
5	Stages where the problem can be detected	CAI , Roller & Squeak & Rattle stage
6	Processes under consideration (boundary/scope of project)	M&M Chassis line & final line, Supplier process and M&M Assembly process
7	Response type (Y):Variable/Attribute.	Attribute
8	Specification of the response:	WSS parameter fail in roller – RBT result fail ; ABS lamp on .
9	Least count of Measurement System (if applicable)	NA
10	Whether MSA is needed? Yes/No	No

1.3 Form Team (involvement Matrix)

Instructions: Use the matrix to help you identify the people who should be involved in identifying cause, developing and implementing solutions.

Task	Details			
	Responsible for	Involved in	Consulted with	Informed About
Identifying causes	CFT	R&D & SQE	Supplier	Supplier
Validating the causes	CAI & MQA			
Identifying Solutions	CAI ,MQA & R&D			
Selecting Solutions	CAI, MQA , R&D & Supplier	MQA, R&D & SQE		Supplier
Planning Implementation	R&D	R&D	Supplier	
Handling potential problems	MQA & R&D	R&D	Supplier	
Implementing solutions	R&D			
Monitoring results	MQA			CQA

Do we need involve anymore person in team? No

1.4 Project Plan

Project Planning

QC story Steps	Mar-19				April- 19,May- 19				Jun-19 , Jul-19				Aug-19				Sep-19			
	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4
1.Select the Theme																				
2.Justify the choice & Containment action																				
3.Understand the current situation																				
4.Select the target																				
5.Analyze																				
6.Implement the corrective measures																				
7.Verify the results and Effects																				
8.Standardize																				
9.Summarize and Future actions	Plan								Actual											

Plan

Actual

2 Justify the choice and containment action

2.1 Past trend of the problem
2.2 Pareto Diagram
2.3 Impact of the problem
2.4 Cost of Poor Quality
2.5 Containment action

Linkage with sector level goal

Crusade 2.0

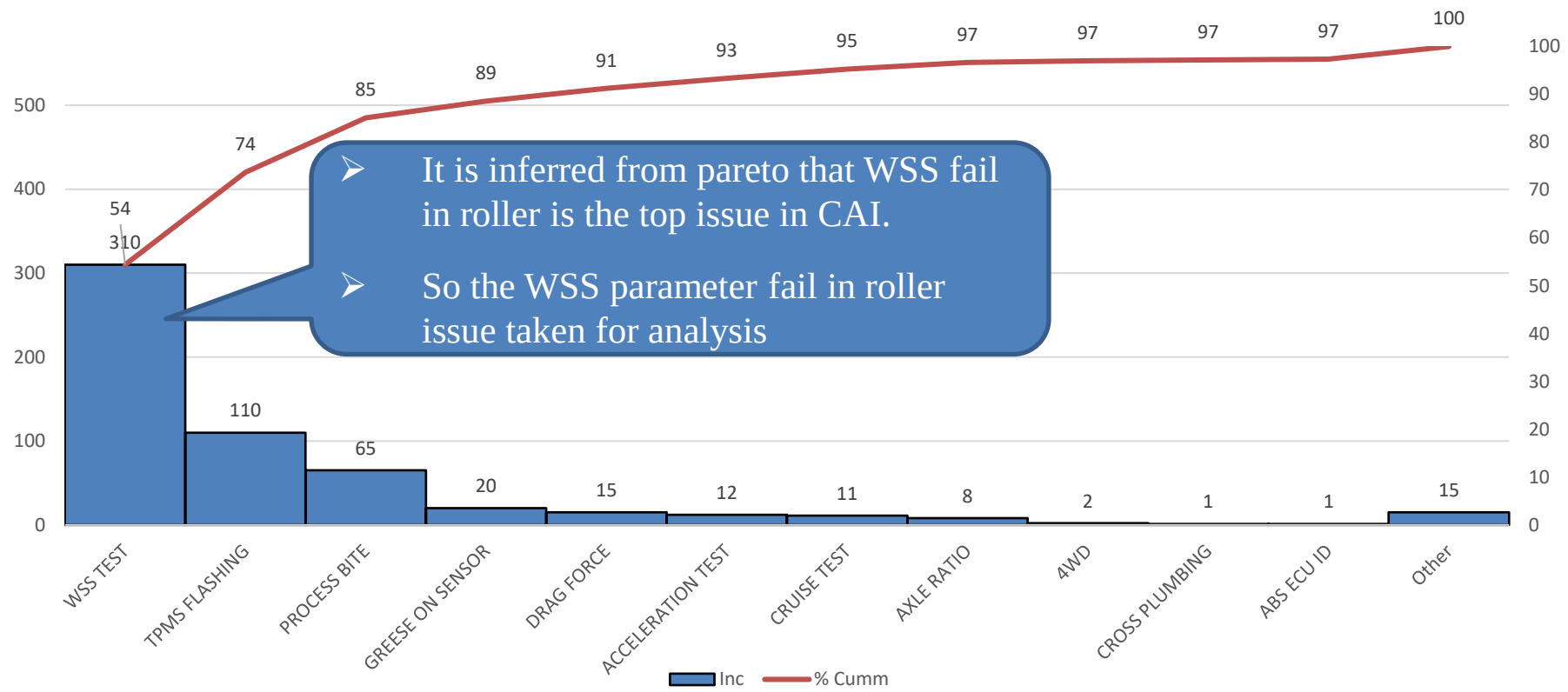
Departmental Objective :-
– DPR Improvement
- Offline rework reduction

Linkage with sector level goal

1. Tree Diagram

2.2 Pareto Diagram

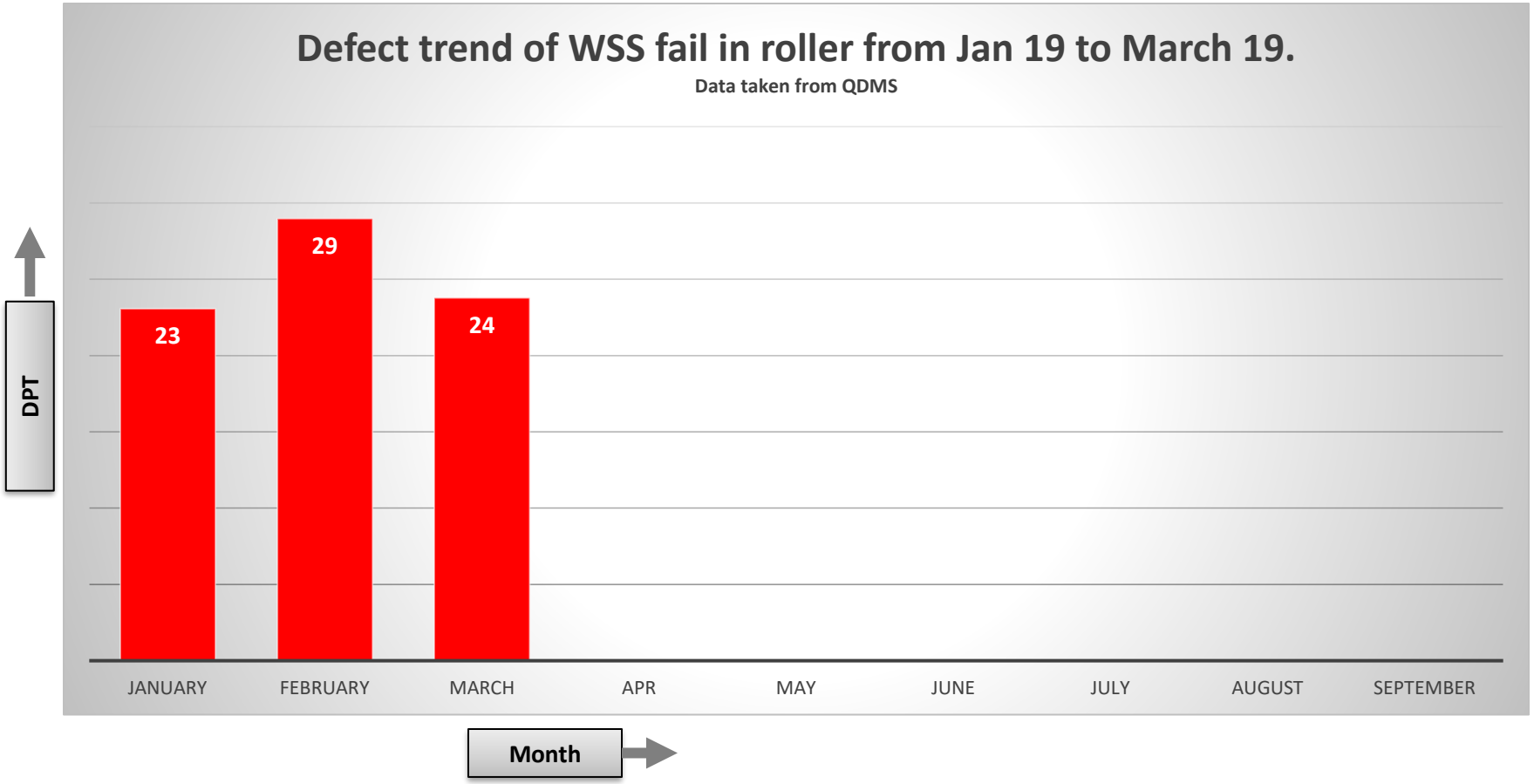
CAI Roller test area defect data trend from Jan 19 to March 19



- It is inferred from pareto that WSS fail in roller is the top issue in CAI.
- So the WSS parameter fail in roller issue taken for analysis

Inference – Highest defect % contributing concern taken as project for resolution.

2.1 Past Trend of the problem



Inference: Defect contribution month to month is consistent

2.3 Impact of the problem

Impact on customer

1. Major functional issue – V1+ , ABS system malfunction.
2. Delay in Delivery of vehicle to RFD with in 1day
3. Customer Dissatisfaction & irritation due to abs lamp on.

Impact on Mahindra brand

4. Customer Delightful experience with Mahindra Brand affected
5. Promise 2022 affected.

Impact on Department goal

1. Major field failure if appear in field - 0 MIS score.
2. Quality indicators affected – FRC / STR
3. Rework cost increased
4. Rework time affected
5. Delivery of the vehicle affected
6. RTR fire

2.4 Cost of Poor Quality

Rework Cost estimation		
Description	Units	
Rework time (Approx)	Min	85
Avg Concerns/Day	Nos	5
Total Required Rework time per day	Min	425
Effective working time	Min	465
Operator required /day	nos	0.91
Approx Operator cost/ day	Rs	325
Operator cost /concern	Rs	297.04
Avg no. of working days	days	25
Operator cost /month	Rs	7426.08
Part reject/month	Nos	3.00
Part Cost	Rs	0.00
Operator Cost/year	**Rs**	**89112.90**

2.4 Cost of Poor Quality

Summary of total benefits expected from the project:

- Improvement in Quality indicators
  (DPT Reduction, STR improvement)
- Cost saving of **Rs.** 89112 **/-** per year – as per COPQ
- Rework time saving of 85 min per vehicle.
- Rework manpower saving of 0.9 persons per day.
- Quality Crusade targets improvement by First time right ,Every time right & Customer delight.
- 6. Vehicles passing to RFD within 1 day achieved.

2.5 Containment Action

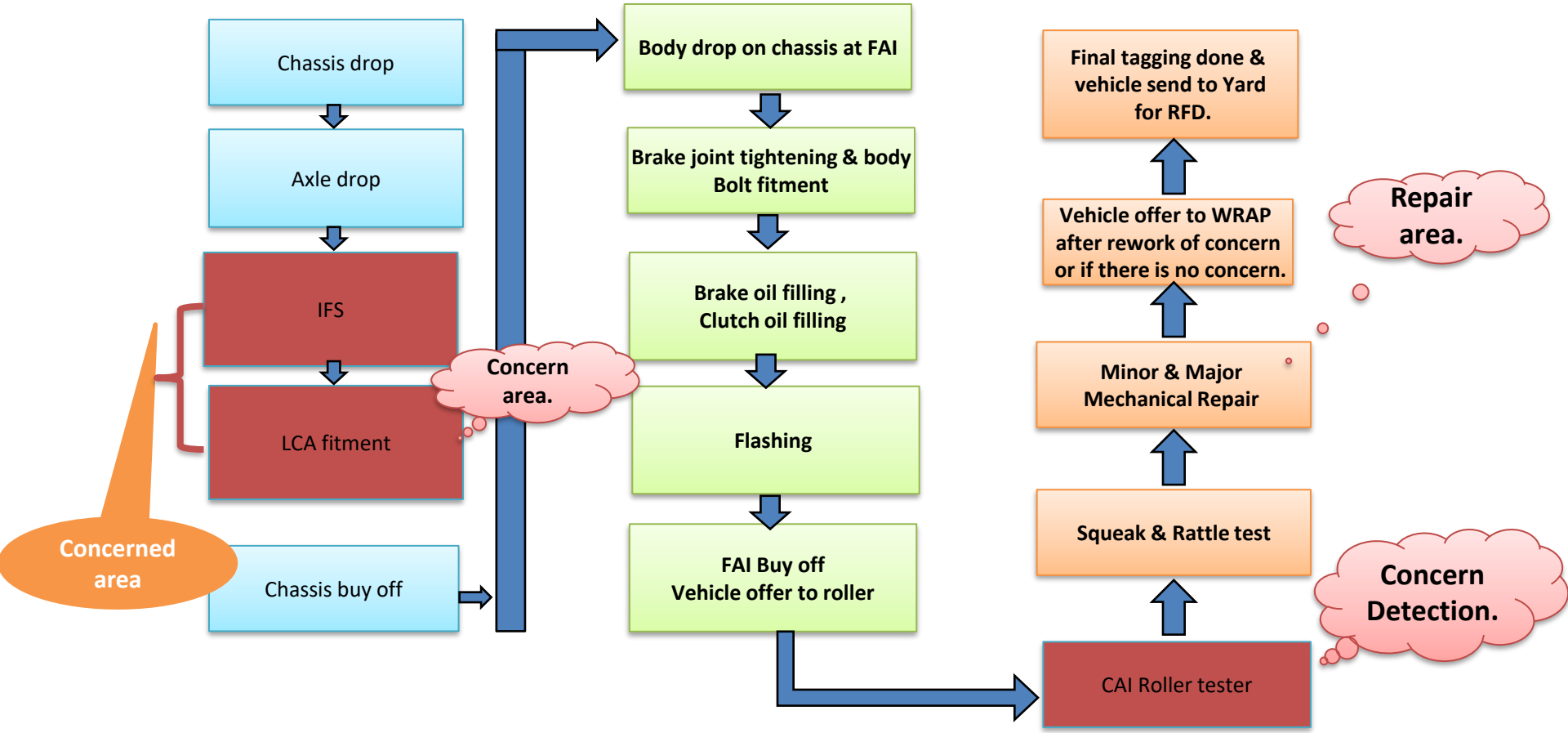
- Awareness given to stage associate and buy off check man
- Awareness given for defect identification & rework at stages of CAI.

3. Understand the current Situation

3.1 Process Flow Diagram
3.2 Past data Analysis from different view point
3.3 Gemba Observations
3.4 Quick win Identification
3.5 Results of Quick wins

3.1 Process Mapping

Process flow



3.2 Different View Point

A. Time to Time Variation:

- Issue reported in both shift.

- Issue reported in month on month no significant difference in occurrence.

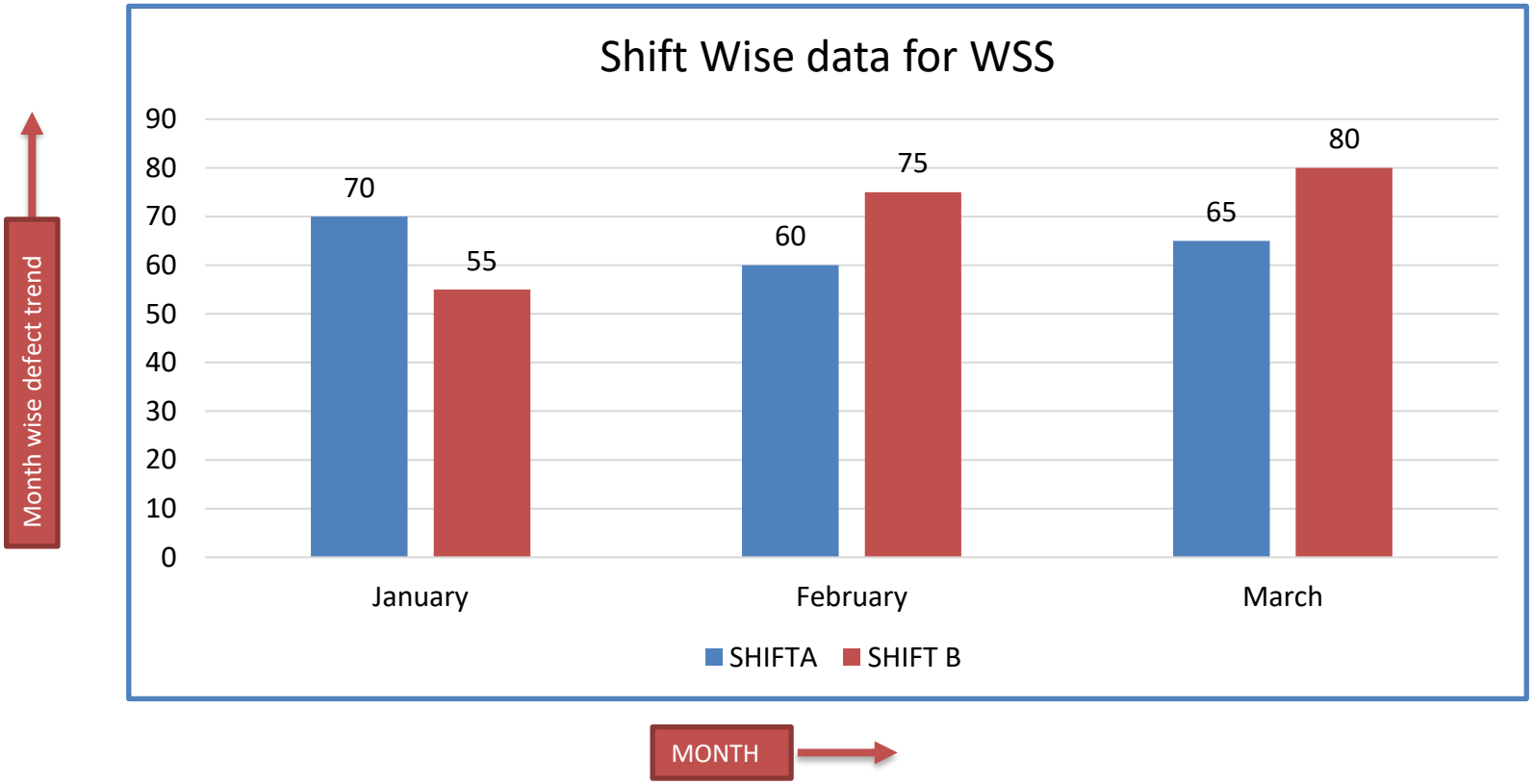
B. Stream to Stream

- Issue reported in Scorpio all variants
- Issue not reported in Marazzo & Xylo model.

Inference :- No significant relation established.

3.2 Different View Point

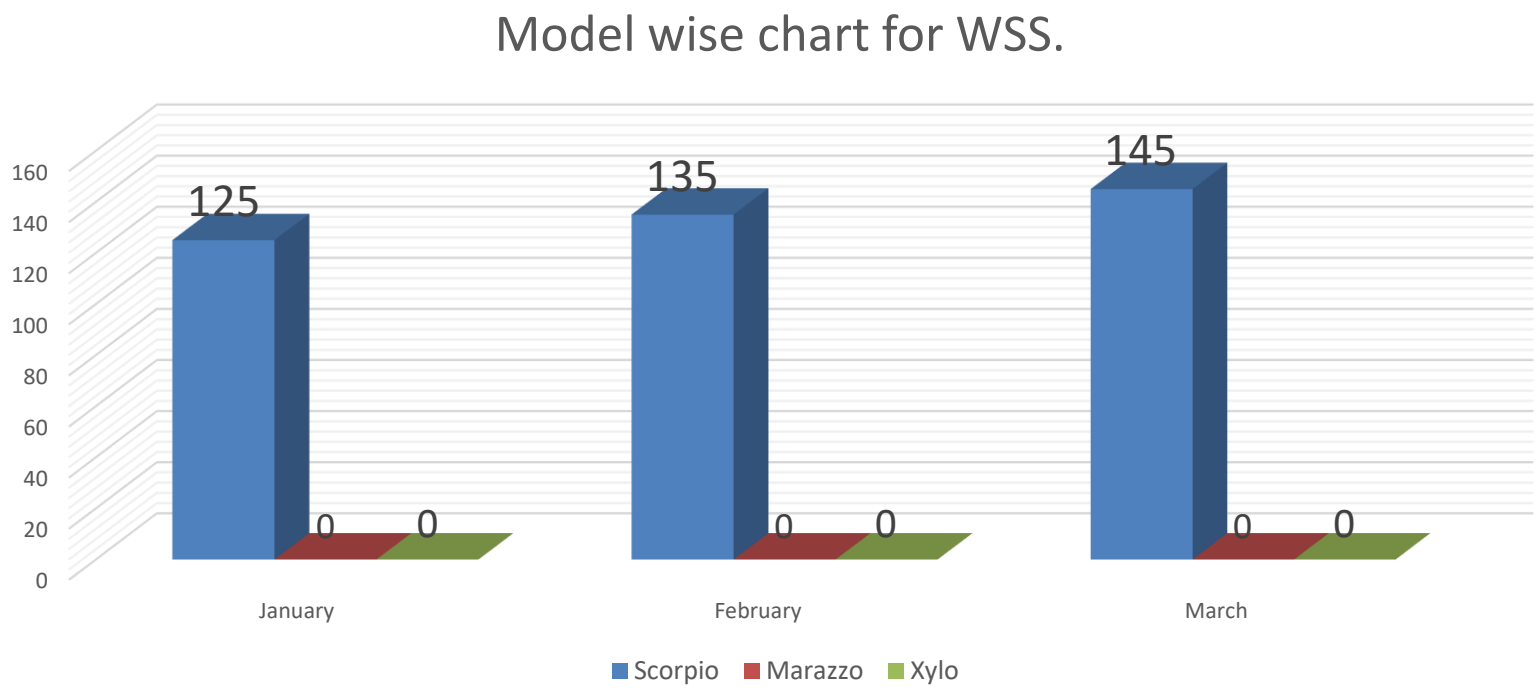
A. Graph of Time to Time Variation:



Inference :- No significant relation established as the issue observed in both the shifts.

3.2 Different View Point

B. Stream to Stream



Inference :-

- Issue observed in only Scorpio model .

3.3 Gemba Observations

- **During Gemba below issues were observed**

1. OBD cable found loose , which may result to CAN communication error.
2. No reference for WSS Sensor position .
3. Toe ring found dusty as it may create error to sense position

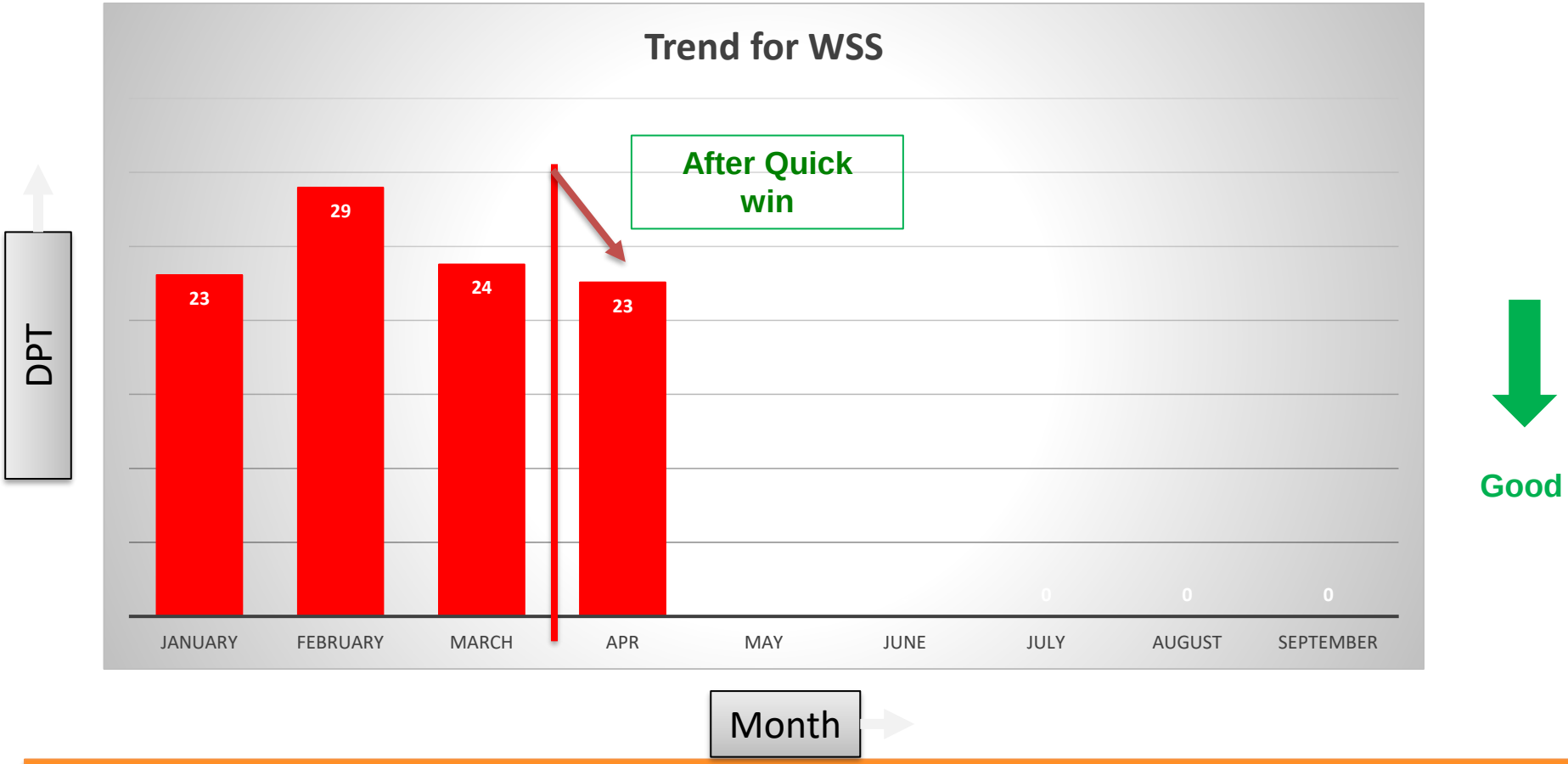
3.4 Gemba observation action plan

Observation	Idea	Source	Easy to implement?	Fast to implement?	Cheap to implement?	Easily Reversible?	Within Authority?	Implement Quick Win
OBD cable found loose , which may result to CAN communication error.	Cable repaired , OBD cable changed.	**Process Audit**	**YES**	**YES**	**NO**	**YES**	**YES**	**YES**
No reference for WSS Sensor position .	Key point & gauge on line for correct position.	**Process Audit**	**YES**	**YES**	**YES**	**YES**	**YES**	**YES**
Toe ring found dusty as it may create error to sense position	As an ICA area Toe ring cleaned & supplied.	**Process Audit**	**YES**	**YES**	**NO**	**YES**	**YES**	**YES**

3.4 Gemba observation countermeasures

Sr No	Observation	Action plan	Responsibility	Target date	Status
1	OBD cable found loose , which may result to CAN communication error.	OBD cable changed , connectivity checked for CAN communication .	**Mahesh Savkar** **Sagar Kekade**	**2.4.19**	**Closed**
2	No reference for WSS Sensor position .	Key point addressed in SOS , awareness given to stage associates.	**Suryabhan Sor** **Mahesh Savkar**	**2.4.19**	**Closed**
3	LCA found dusty as it may create error to sense position .	Awareness given to supplier .	**Satish Pagar** **Santosh Jadhav** **Mahesh Savkar**	**2.4.19**	**Closed**

3.5 Results of Quick wins



Inference:
There is no significant reduction in defect rate of brake switch setting error

3.5 Results of Quick wins

Practical statement: WSS parameter fail in roller defect rate got reduced after taking corrective measures through quick win.

Statistical problem:

Ho: Defect rate after Quick win >= Defect rate before quick win.

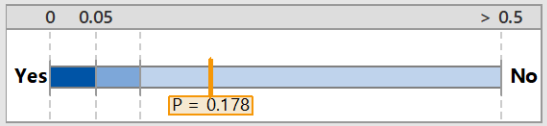
Ha: Defect rate after Quick win < Defect rate before quick win.

Analysis: 2P Test

3.5 Results of Quick wins

2-Sample % Defective Test for Before vs After Quick
Summary Report

% Defective Test
Is Before greater than After Quick ?



The % defective of Before is not significantly greater than the % defective of After Quick (p > 0.05).

Individual Samples

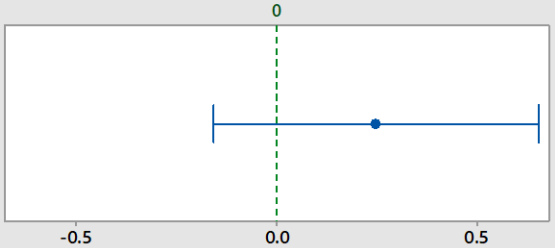
Statistics	Before	After Quick
Total number tested	16175	4831
Number of defectives	405	109
% Defective	2.50	2.26
90% CI	(2.31, 2.72)	(1.92, 2.64)

Difference Between Samples

Statistics	*Difference
Difference	0.25
90% CI	(-0.16, 0.65)

*Difference = Before - After Quick

90% CI for the Difference
Is the entire interval above zero?



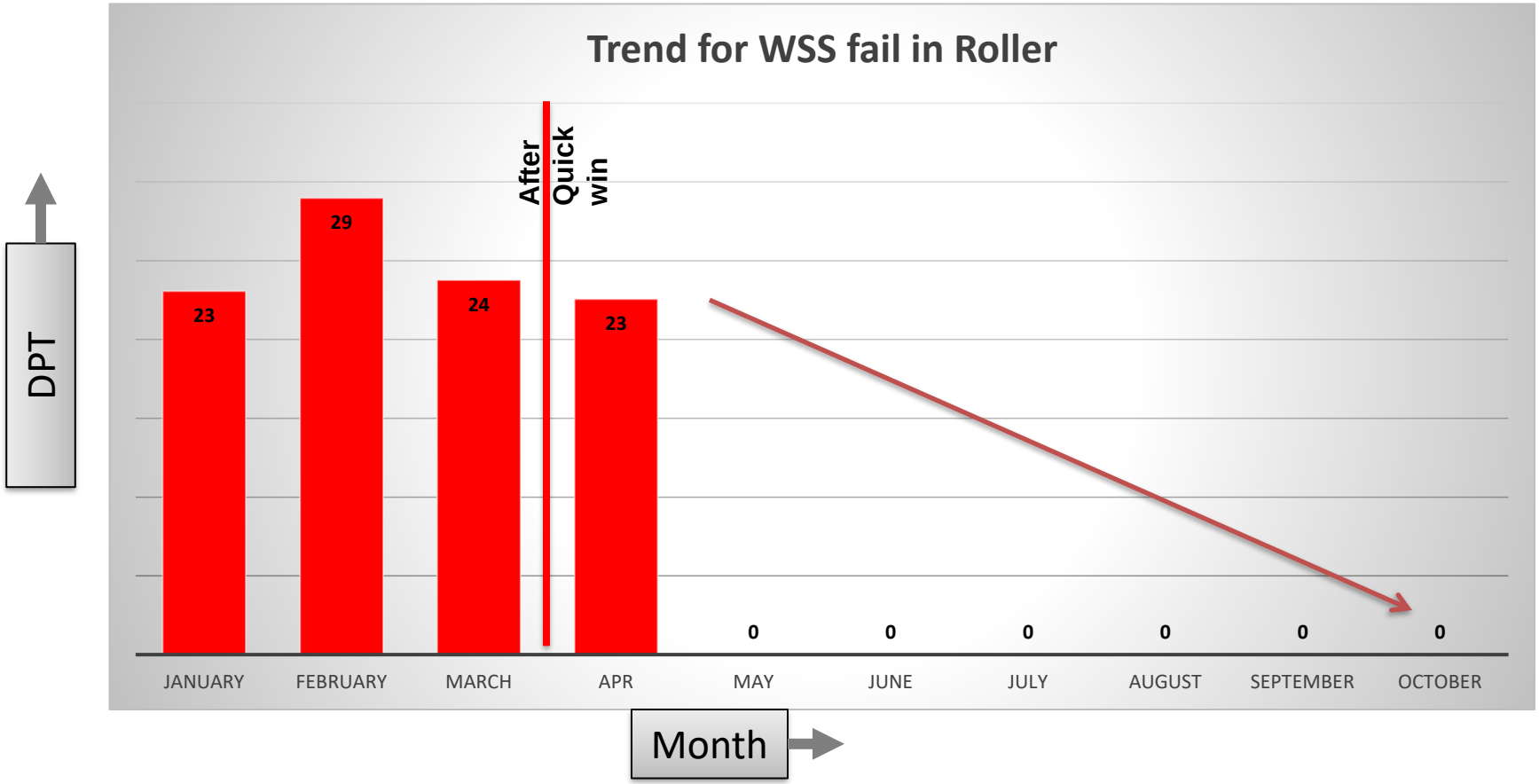
Comments

- Test: There is not enough evidence to conclude that the % defective of Before is greater than After Quick at the 0.05 level of significance.
- CI: Quantifies the uncertainty associated with estimating the difference from sample data. You can be 90% confident that the true difference is between -0.16% and 0.65%, and 95% confident that it is greater than -0.16%.

Statistical Conclusion: We Cannot reject Ho as P Value observed is > 0.05

Practical Conclusion: **Defect rate of <u>WSS parameter fail in roller</u> was not significantly reduced after taking corrective measures through quick win.**

4. Select the target



Current Level: Average 25 DPT at STCF. Affecting the STR and FRC.

Target Level: by 30 th October concern elimination.

5. Analyze

5.1 List the SSVs

5.2 Identify the tool

5.3 Collect the data

5.1 List down SSV's or Xs

$$Y = f(x)$$

Response SSV

Y = WSS parameter fail in Roller

X = X1. Assembly process
X2. Components in assembly

- **Initially CFT decided to find the root cause from Assembly process (X1)**

- **Assembly process consist of –**

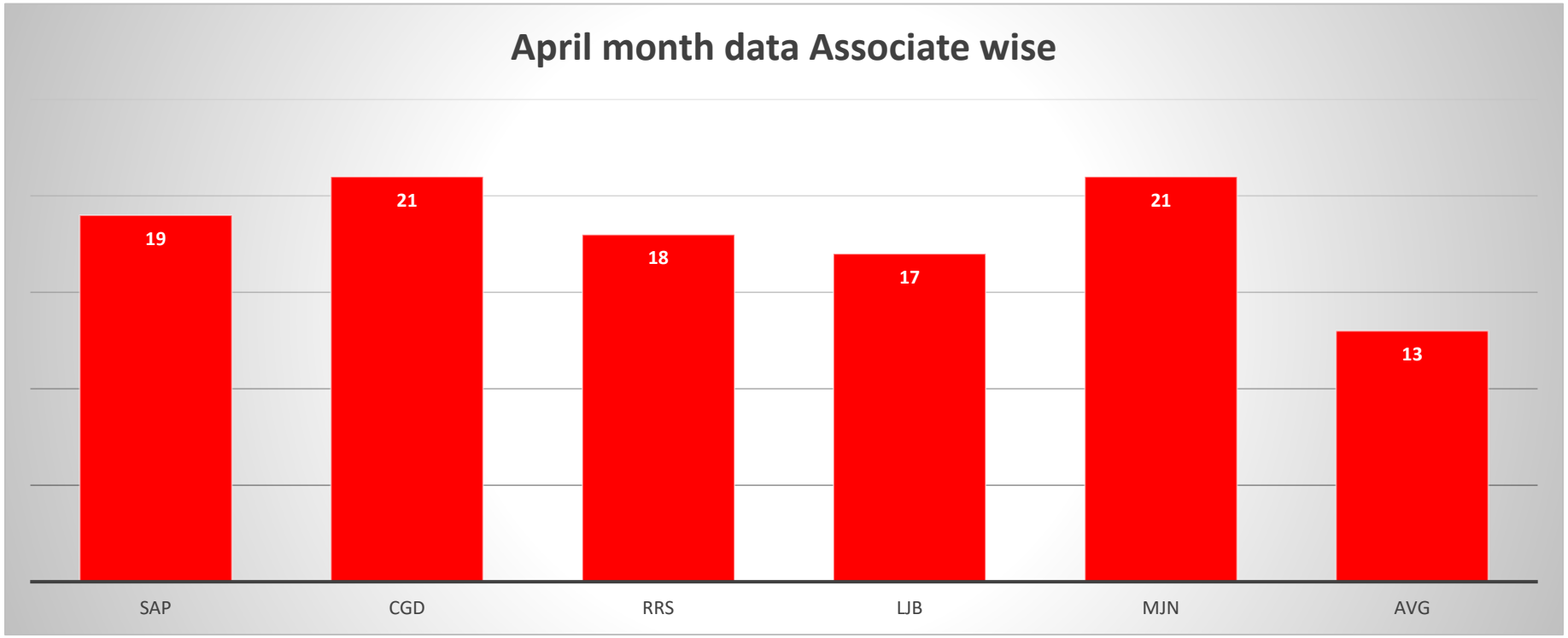
A. Man-Method contribution - chi square test to confirm associate contribution

B. Machine contribution – DOE analysis

5.1 List down SSV's or Xs

A. Man-Method contribution.

To check the man contribution in defect generation we have checked all associate contribution for last one month defect incidences.



Inference : For assembly process , all stage associates data checked for April month , No abnormality observed in process. All associates are Black I. So, X1 not a contributing factor

5.1 List down SSV's or Xs

A. Man-Method contribution.

There are more than two associate checking vehicles – to confirm individual contribution for defect generation Chi-Square test is done.

Practical statement:

WSS test failed Proportion of failure rate is different for different operators

Statistical problem:

H0: Defects (LJB) = Defects (CGD) = Defects (AVG) = Defects (RRS) = Defects (SAP) = Defects (MJN) =

Ha: At least one of the associate is different

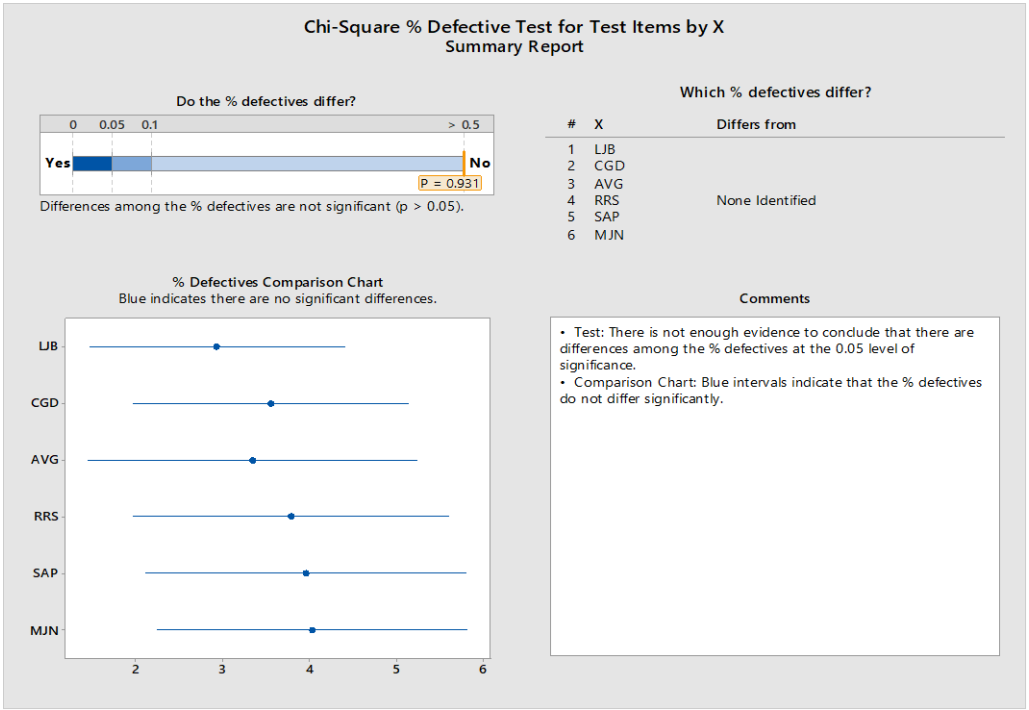
Analysis: 2P Test

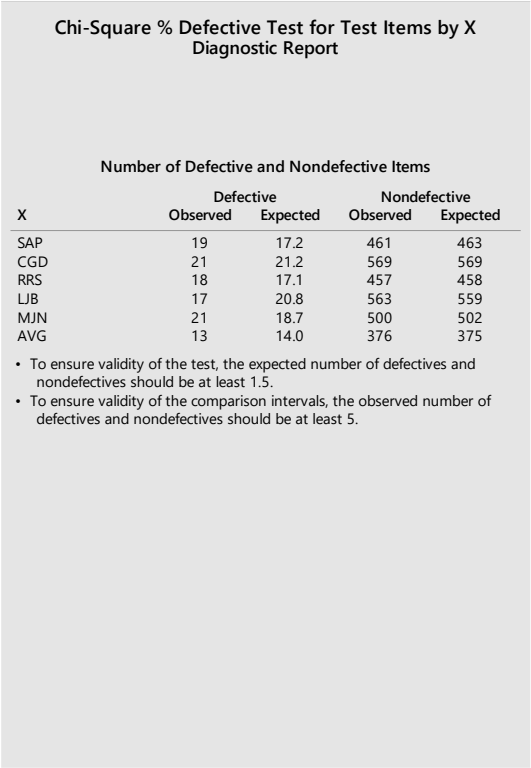
Note – LJB / CGD / AVG / RRS / SAP / MJN ------- Operator Initial names.

5.1 List down SSV's or Xs

A. Man-Method contribution.

There are more than two associate checking vehicles – to confirm individual contribution for defect generation Chi-Square test is done.





Chi-Square % Defective Test for Test Items by X
Diagnostic Report

Number of Defective and Nondefective Items

X	Defective Observed	Defective Expected	Nondefective Observed	Nondefective Expected
SAP	19	17.2	461	463
CGD	21	21.2	569	569
RRS	18	17.1	457	458
LJB	17	20.8	563	559
MJN	21	18.7	500	502
AVG	13	14.0	376	375

• To ensure validity of the test, the expected number of defectives and nondefectives should be at least 1.5.
• To ensure validity of the comparison intervals, the observed number of defectives and nondefectives should be at least 5.

<u>Inference</u> : P value is >0.05 so there is no enough evidence to conclude that this defect is associate specific, No contribution to associate in WSS sensor parameter fail.

5.1 List down SSV's or Xs

X2 : Roller machine

Sr no	SXTCF : Roller Machine	BTCF : Roller machine	Again ReRoller done in SXTCF
1	fail	pass	fail
2	fail	pass	pass
3	fail	fail	fail
4	fail	pass	fail
5	fail	pass	pass

Inference : 5 vehicles roller fail in SXTCF Roller machine, same vehicles roller done in BTCF . In this 4 vehicles roller passed. Again same 5 vehicles Reroller done SXTCF roller machine 3 vehicles failed, & 2 passed. Which means problem belongs to Roller machine.

5.1 List down SSV's or Xs

- Team listed the parts in descending order of roller machine, in which it will affect the response

Sr. No	Roller Part involve for RBT test WSS failure mode.
1	Roller Machine Encoder at LH and RH location
IMAGES	

Encoder

A **encoder**, is an device that converts the motion to digital output signals.

An **encoder** generates a pulse for each **incremental** step in it's rotation. The disk is rotating in a counter-clockwise direction. Therefore, by monitoring both the number of pulses and the relative phase of signals of wss, you can track both the position and direction of rotation.

5.2 Identify the tool

- Team listed the parts of Roller in descending order in which will affect the response

Sr. No	Part Description	Usage of parts after disassembly
1	LH ENCODER	**Can be reused**
2	RH ENCODER	**Can be reused**

Hence DOE Tool

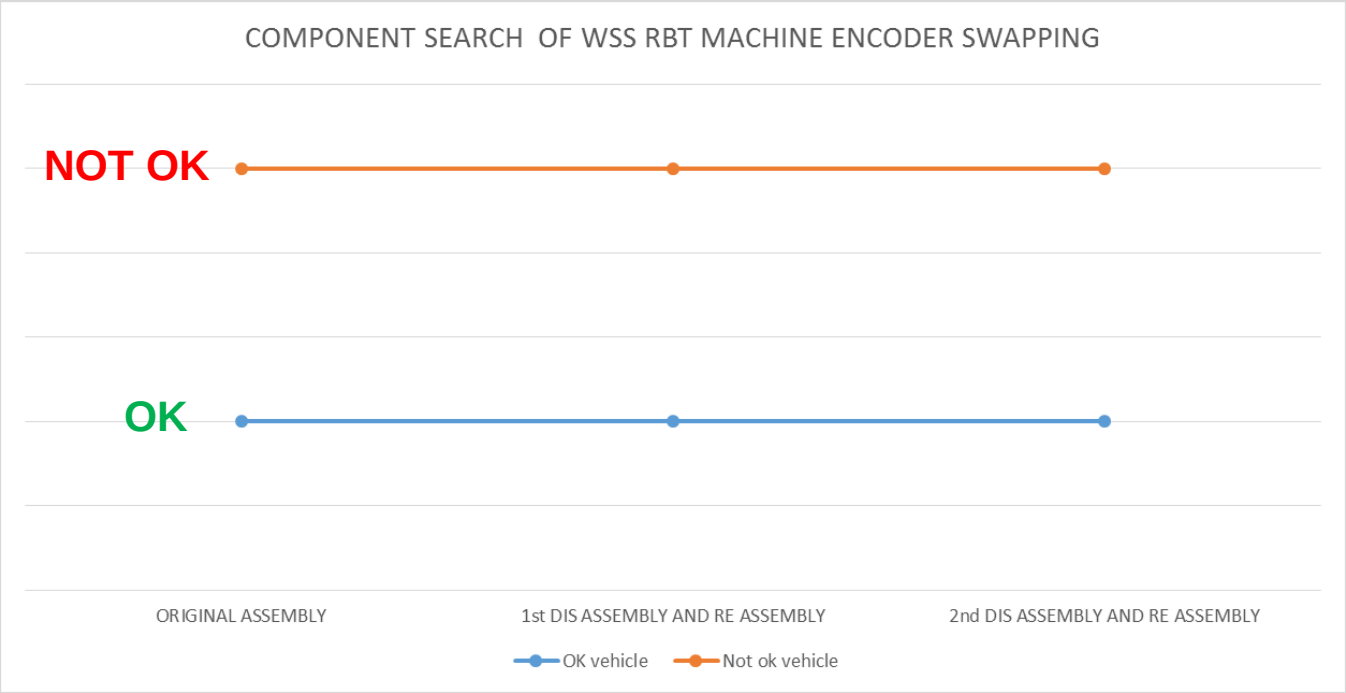
"COMPONENT SEARCH"

was selected

5.2 Analysis

Shainin DOE - Component Search – Graphical Presentation :-

A) Assembly Process Confirmation



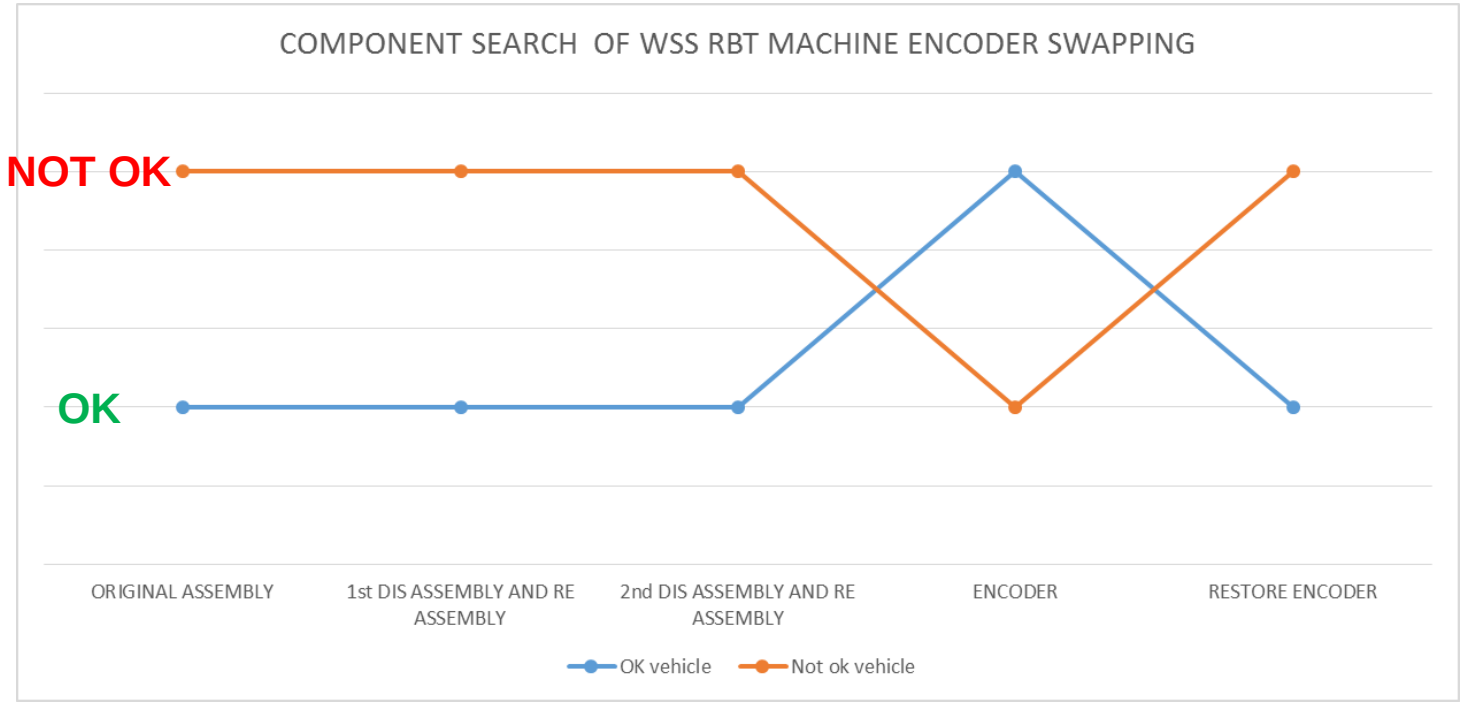
Inference :- Assembly Process is not a cause.

While assembly and dis assembly of encoder – defective vehicle observed defective and ok vehicle observed ok – No in version of defect.

5.2 Analysis

Shainin DOE - Component Search – Graphical Presentation :-

A) Component from RBT test –ENCODER -



Inference :- ENCODER, component from RBT machine is a cause.

While RH/LH ENCODER swapping – defective vehicle OK and ok vehicle defective – Complete in version of defect.

5.6 Funneling to Next level causes

Shining DOE confirms Problem in M/C encoder.

Roller tester machine Encoder found defective

Why ?

Encoder inner shaft found defective

Why ?

Shaft found wear out and it is not gets identified

Why ?

Shaft wear out verification not done

Why ?

This point was not addressed in M/C PM check sheet.

HOW`

Point to be added in M/C PM check sheet.

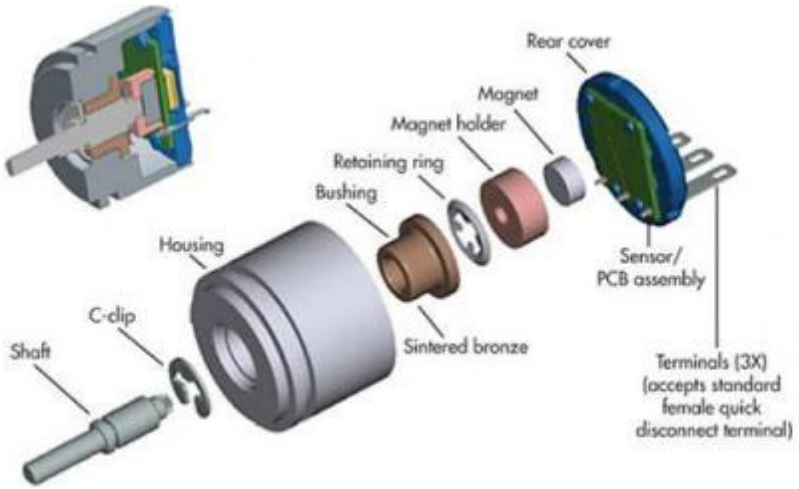
ROOT CAUSE

6. Implement Corrective Measures

1. Generate Possible Solutions
2. Evaluate and Select Solution
3. Implement the action

6.1 Generate possible solutions

X	Validated causes	Possible Solutions
X1	Encoder Malfunction	DOE done of encoder, Encoder shaft found wear out . Encoder to be changed.



This shaft wear out was the natural deterioration , which was not got identified during routine machine PM as it was not added in PM check sheet.

6.2 Evaluate and Select Solution

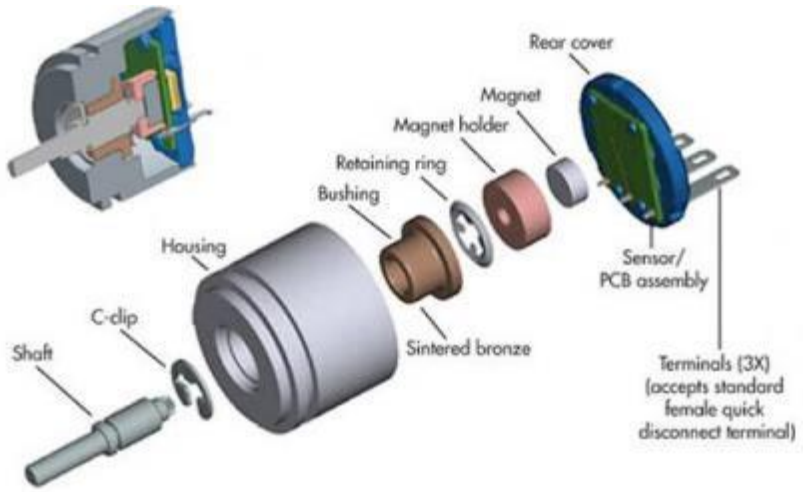
Solution Selection Matrix

	Effectiveness	Time Impact	Cost/Benefit Impact	<<<<Evaluation Criteria
	10	8	5	<<<<<<<<Importance
Solution	Correlation of Solution to Criteria			Total
Encoder to be changed.	10	9	10	222

6.3 Implement Solution

Solution Implementation Plan

X	Validated causes	Actions finalized	Resp.	Target Date	Status
X1	Encoder Shaft wear out	New Encoder fitted. Point added in PM check sheet.	Sagar KEKADE	22.05.19	IMPLEMENTED

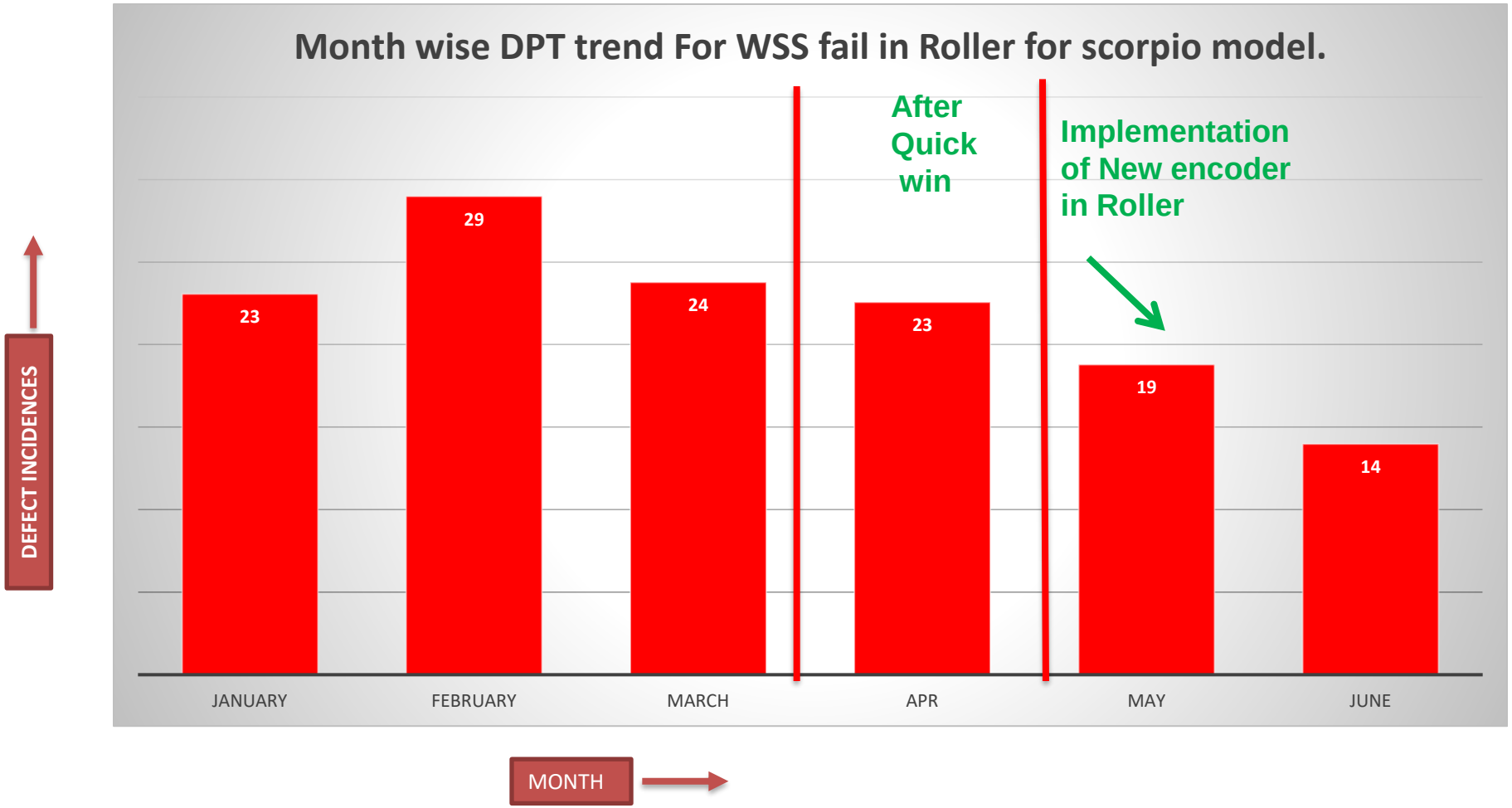


7. Confirm Results and effects

1. Collect data
2. Capture all effects
3. Compare before and after
4. Measure Benefits

7.1 Collect data

Data of after action



Inference: **It means defect rate of WSS fail in roller was reduced but not 0 down.**

7.2 Capture all the effects

Effects on the selected problem:

- FRC improved partially.
- Concern defect rate reduced but not 0 down
- STR loss improved
- Rework time saved partially
- Rework manpower utilized for other reworks
- Vehicles passing to RFD within 1 day improved

Negative effects observed:

- NIL

7.3 Compare Before and After

Hypothesis testing for confirmation of results after PCA actions taken :-

Practical statement –

Defect rate of WSS fail in Roller (Y) was reduced after taking corrective measures through encoder changed.

Statistical Problem –

Ho: Defect rate after corrective measure >= Defect rate before corrective measure.

Ha: Defect rate after corrective measure < Defect rate before corrective measure .

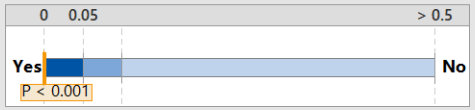
Analysis: 2P Test -

7.3 Compare Before and After

Analysis: 2P Test

2-Sample % Defective Test for Before vs After encode
Summary Report

% Defective Test
Is Before greater than After encode?



The % defective of Before is significantly greater than the % defective of After encode (p < 0.05).

Individual Samples

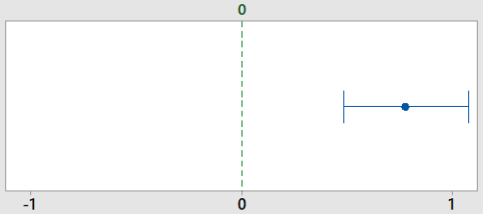
Statistics	Before	After encode
Total number tested	21006	7835
Number of defectives	514	131
% Defective	2.45	1.67
90% CI	(2.27, 2.63)	(1.44, 1.93)

Difference Between Samples

Statistics	*Difference
Difference	0.77
90% CI	(0.48, 1.07)

*Difference = Before - After encode

90% CI for the Difference
Is the entire interval above zero?



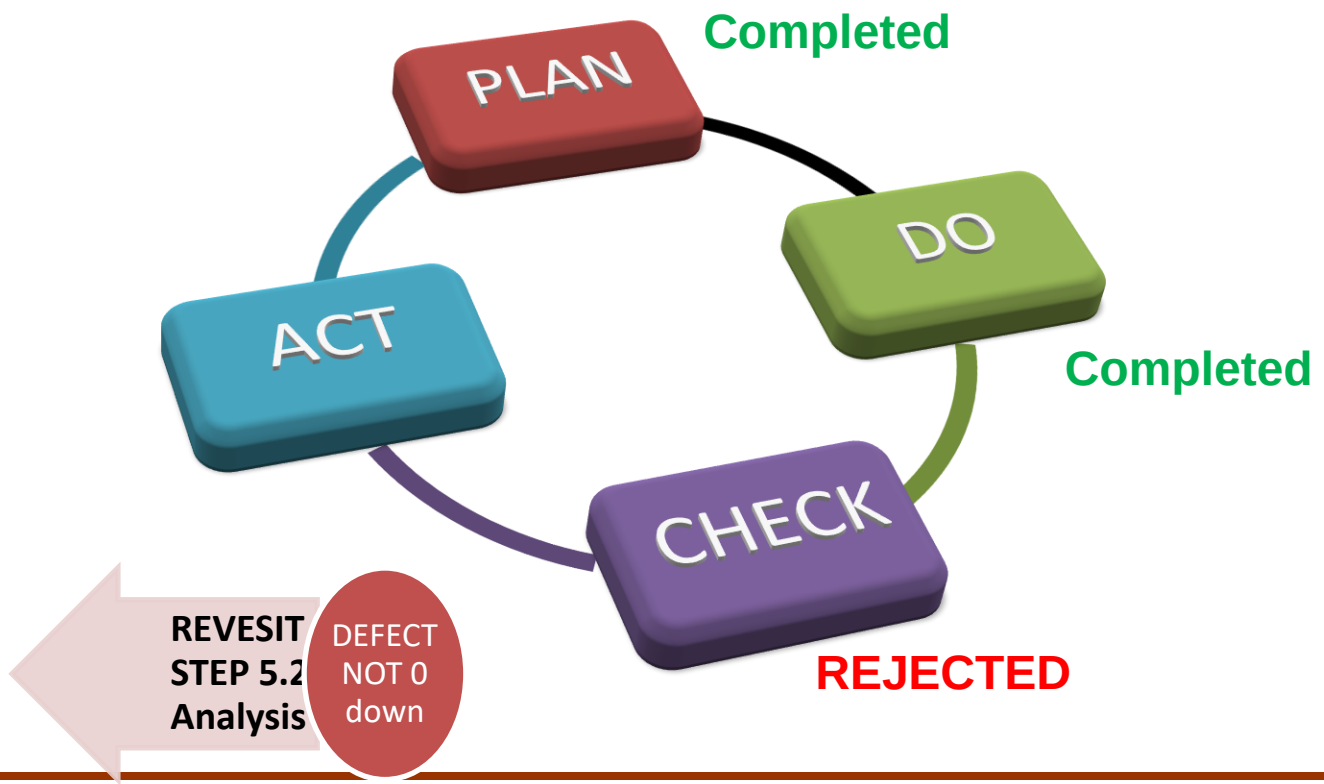
Comments

- Test: You can conclude that the % defective of Before is greater than After encode at the 0.05 level of significance.
- CI: Quantifies the uncertainty associated with estimating the difference from sample data. You can be 90% confident that the true difference is between 0.48% and 1.07%, and 95% confident that it is greater than 0.48%.

Statistical Conclusion:
Reject the null hypothesis as P Value observed is < 0.05
Practical Conclusion:
Defect rate of WSS parameter fail (Y) was reduced after taking corrective actions through Encoder changing.

7.3 Compare Before and After

Concern actions was giving effect to reduce defect but not making concern 0
So need to continue the SHAININ DOE analysis



5.1 List down SSV's or Xs

$$Y = f(x)$$

Response SSV

Y = Component is the problem

X1= LCA
X2= WSS sensor

5.2 Identify the tool

- Team listed the parts of Roller in descending order in which will affect the response

Sr. No	Part Description	Usage of parts after disassembly
1	LCA	**Can be reused**
2	WSS SENSOR	**Can be reused**

Hence DOE Tool

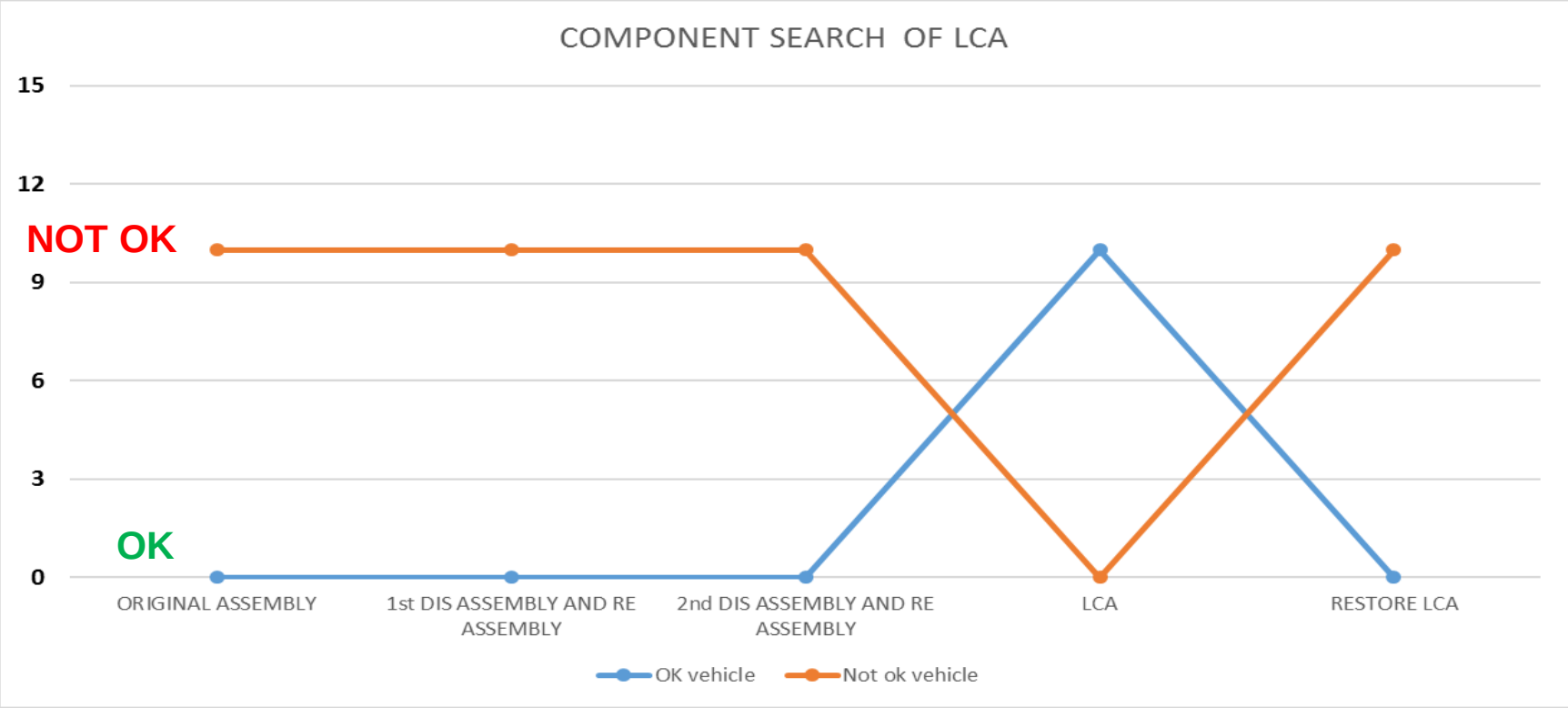
"COMPONENT SEARCH"

was selected

5.2 Analysis

Shainin DOE - Component Search – Graphical Presentation :-

A) Assembly process confirmation of components.



Inference :- LCA is a cause.

While LCA swapping – defective vehicle OK and ok vehicle defective – **Complete in version of defect.**

5.3 Collect the data

SR NO	VIN	MODEL	ROLLER RESULT	BURR OBSERVED ON TOE RING
1	12125	S3	FAIL	YES
2	12183	S11	FAIL	YES
3	12283	S11	FAIL	YES
4	12187	S3	FAIL	YES
5	12483	S11	FAIL	YES
6	12854	S3	FAIL	YES
7	11173	S5	FAIL	YES
8	11213	S3	FAIL	YES
9	11346	S9	FAIL	YES
10	11420	S9	FAIL	YES

Inference :- 10 vehicles data collected , all vehicles roller fail in wss observed Burr on toe ring.

5.4 Funneling to Next level causes

Shining DOE confirms problem lies with Cargo assembly.

WHY
- SHININ DOE confirms that defect follows the LCA.

WHY
- WSS sensor counts the rotation .

WHY
- Burr on Toe ring disrupts the operation of count.

WHY
- Supplier don't have process control for this Foreign particle entrance on Toe ring

HOW
- Protective cap to be provided on Toe ring.

CONCERN TAKEN UP WITH SUPPLIRE FOR PROTECTIVE CAP ON TOE RING

COUNTER MEASURE

6. Implement Corrective Measures

1. Generate Possible Solutions
2. Evaluate and Select Solution
3. Implement the action

6.1 Generate possible solutions

X	Validated causes	Possible Solutions
X1	Burr on Toe ring .	Cover plate to be added on Toe ring to avoid metal burr particles entrance on Toe ring.

BURR ENTRANCE ON TOE RING DUE TO NO COVER PLATE

COVER PLATE ON TOE RING TO AVOID FOREIGN PARTICLE ON TOE RING.

6.2 Evaluate and Select Solution

Solution Selection Matrix

	Effectiveness	Time Impact	Cost/Benefit Impact	<<<<Evaluation Criteria
	10	8	5	<<<<<<<<Importance
Solution	Correlation of Solution to Criteria			Total
Cover plate provision on Toe ring.	10	8	10	194

6.3 Implement Solution

Solution Implementation Plan

X	Validated causes	Actions finalized	Resp.	Target Date	Status
X1	Burr on Toe ring	Cover plate to be added on Toe ring to avoid metal burr particles entrance on Toe ring.	Prashant Tarte.	17th July	IMPLEMENTED

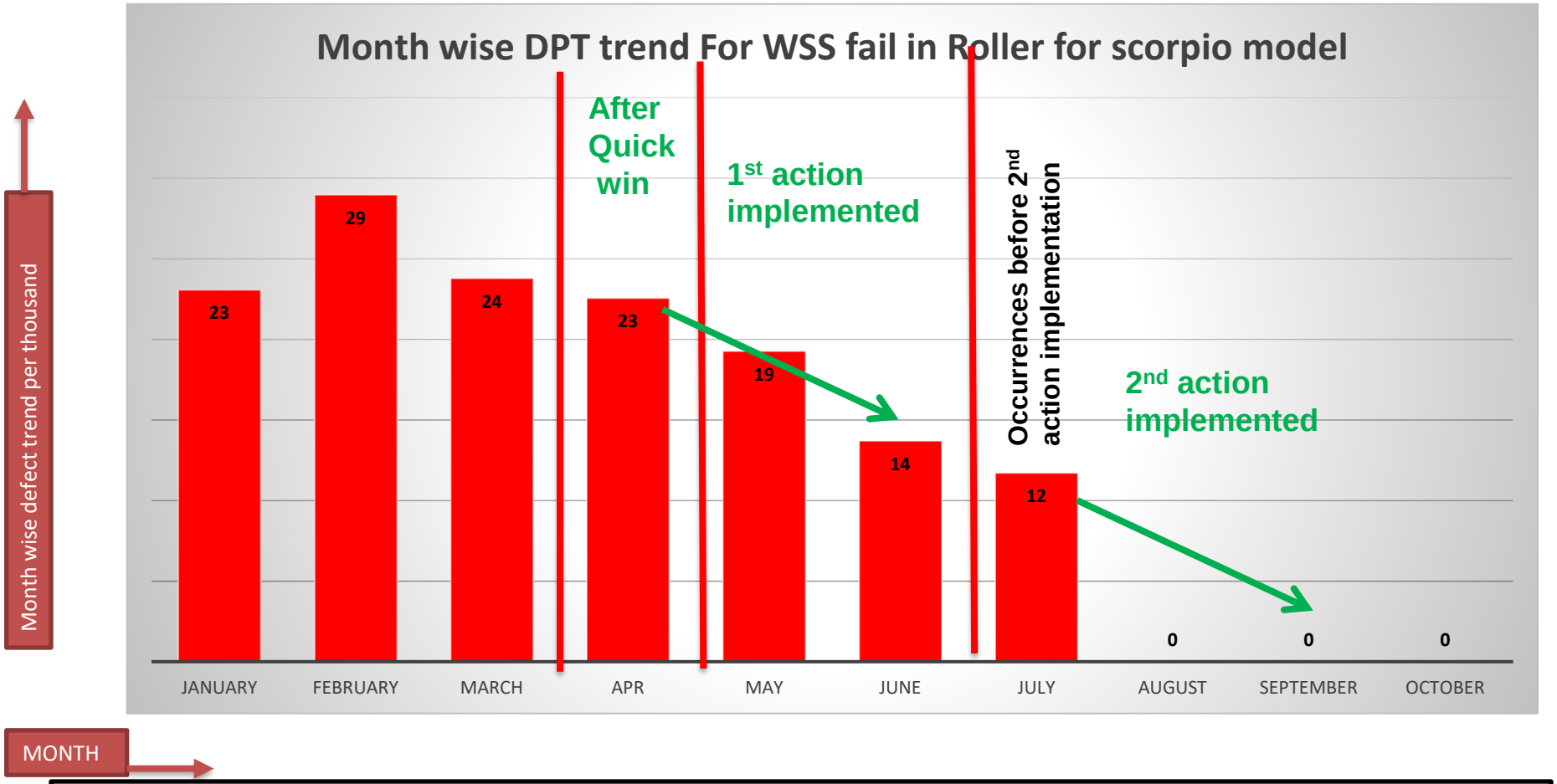
COVER PLATE ON TOE RING TO AVOID FOREIGN PARTICLE ON TOE RING.

7. Confirm Results and effects

1. Collect data
2. Capture all effects
3. Compare before and after
4. Measure Benefits

7.1 Collect data

Data of after action – Month wise defect trend



Inference: Concern is totally eliminated after taking 2nd action.

7.2 Capture all the effects

Effects on the selected problem:

- FRC /FBO improved.
- First time right achieved
- STR improved from 70 to 70.5
- Rework time saved
- Rework manpower utilized for other reworks
- Vehicles passing to RFD within 1 day achieved

Any other positive effects observed:

- RFI defect eliminated.
- Customer delight about Scorpio will increase.

Negative effects observed:

- NIL

7.3 Compare Before and After

Hypothesis testing for confirmation of results after PCA actions taken :-

Practical statement –

Defect rate of Roller test fail in WSS (Y) was eliminated after taking corrective measures through correction actions.

Statistical Problem –

Ho: Defect rate after corrective measure >= Defect rate before corrective measure.

Ha: Defect rate after corrective measure < Defect rate before corrective measure .

Analysis: 2P Test -

7.3 Compare Before and After

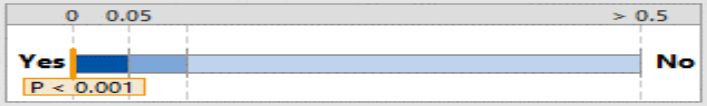
Analysis: 2P Test

2-Sample % Defective Test for Group 1 vs Group 2

Summary Report

% Defective Test

Is Group 1 greater than Group 2?



The % defective of Group 1 is significantly greater than the % defective of Group 2 (p < 0.05).

Individual Samples

Statistics	Group 1	Group 2
Total number tested	7835	10993
Number of defectives	91	0
% Defective	1.16	0.00
90% CI	(0.97, 1.38)	(0.00, 0.02)

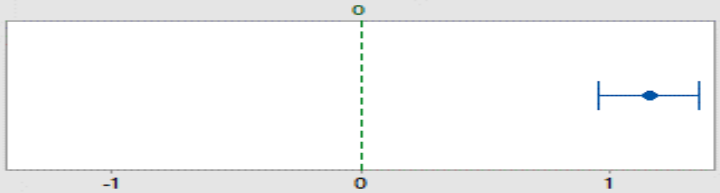
Difference Between Samples

Statistics	*Difference
Difference	1.16
90% CI	(0.96, 1.36)

*Difference = Group 1 - Group 2

90% CI for the Difference

Is the entire interval above zero?



Comments

- Test: You can conclude that the % defective of Group 1 is greater than Group 2 at the 0.05 level of significance.
- CI: Quantifies the uncertainty associated with estimating the difference from sample data. You can be 90% confident that the true difference is between 0.96% and 1.36%, and 95% confident that it is greater than 0.96%.

Statistical Conclusion:

Reject the null hypothesis as P Value observed is < 0.05

Practical Conclusion:

Defect rate of WSS parameter fail in roller (Y) was eliminated after taking corrective actions at LCA by adding cover plate.

7.4 Measure Benefits

Tangible Benefits:

- Cost of Poor Quality saved Rs . 89112
- STR improved from 70 to 70.5 %
- RFD achieved to 96 %.

MAIL FROM INDUSTRIAL ENGG FOR COPQ

From: DHANOKAR RAJESH <DHANOKAR.RAJESH@mahindra.com>
Sent: 04 December 2019 09:45
To: SAVKAR MAHESH <SAVKAR.MAHESH@mahindra.com>; DHANOKAR RAJESH <DHANOKAR.RAJESH@mahindra.com>
Subject: Re:MGB COPQ COST APPROVAL

okay approved

Sent from my Redmi Note 4
On SAVKAR MAHESH <SAVKAR.MAHESH@mahindra.com>, 03-Dec-2019 10:31 PM wrote:
Dear Sir,

In WSS Roller fail issue Below is the rework cost estimation. Kindly approve .

Rework Cost estimation		
Description	Units	
Rework time (Approx)	Min	85
Avg Concerns/Day	Nos	5
Total Required Rework time per day	Min	425
Effective working time	Min	465
Operator required /day	nos	0.91
Approx Operator cost/ day	Rs	325
Operator cost /concern	Rs	297.04
Avg no. of working days	days	25
Operator cost /month	Rs	7426.08
Part reject/month	Nos	3.00
Part Cost	Rs	0.00
Operator Cost/year	**Rs**	**89112.90**

Regards
Mahesh.

Intangible Benefits:

- Loss of STR avoided.
- Customer satisfaction (RFI , ANOVAC & Field)
- Quality indicators improvement.
- Through put time improved.
- Line flow become smooth due to no rework

8. Standardize

1. Update Documents
2. Train people to follow new standards
3. Audit standards

8.1 Update Document

Jump Seat Rattle Noise (Y1):

S N	Document	Target date	Responsibility	Status
1	PM Check sheet updated for encoder.	22.05.19	Sagar Kekade	Closed
2	Drawing updation	24.6.19	Prashant Tarte	Closed
3	EN Release	17.7.19	Prashant Tarte	Closed

8.1 Update Document

PM CHECK SHEET

Preventive Maintenance Sheet - Offline						Frequency- Monthly
Area :- Test Line		Name of Machine :- Roll & Brake Test (Saturday)				Mechanical
Sr.No	Item (What)	Activity	Method (How)	Required Value	C/NC	Remark
1	**RH Side** All pneumatic Cylinders (Retaining Roller-3 , Shot Pin-4, stop post-1 , Exaust-1, Brake-2) **(Ref b.d. 20072990)/(Ref. Bd. 20080452)**	Inspect	With cloth / Visually	Ensure cleaning inside / No air leakage / clean all cylincers / clean filters /No seal damaged/ Ensure all cylinders are coupled to load properly (check lock nut for tightening)/clean shot pins/sensor mounting/cleaning./Cylinder mounting bolts & cylinder end plate bolts should be as per marking./	○	
2	**RH Side** Main Roller Motors - -2 nos & blower motors-2	Inspect	Visuallly /clean / lubricati on	Mounting / cable termination OK / No abnormal noise during running / properly coupled to Load / Fan motor filter cleaning / Motor bearing lubrication/	○	
3	**RH Side** Wedge brake assemblies	Inspect / clean	visually	running freely / no layer of impurities / no excess ware / spring is properly tensioned / Gap between roller & linear is OK	○	
4	**RH Side** Main Rollers	Inspect / clean	Visually	The mouniting of Rollers Ok / bearings Ok / Lubricate bearing / clean roller from inside / balancing of roller	○	
5	**LH Side** Main Roller Motors - -2 nos & blower motors-2	Inspect	Visuallly /clean / lubricati on	Mounting / cable termination OK / No abnormal noise during running / properly coupled to Load / Fan motor filter cleaning / Motor bearing lubrication.	○	
6	**LH Side** All pneumatic Cylinders (Retaining Roller-3 , Shot Pin-4, stop post-1 , Brake-2) **(Ref b.d. 20072990)/(Ref. Bd. 20080452)**		With cloth / Visually	Ensure cleaning inside / No air leakage / clean all cylincers / clean filters /No seal damaged/ Ensure all cylinders are coupled to load properly (check lock nut for tightening)/clean shot pins.sensor mounting/cleaning.	○	
7	**LH Side** Main Rollers	Inspect / clean	Visually	The mouniting of Rollers Ok / bearings Ok / Lubricate bearing / clean roller from inside / balancing of roller	○	
8	**LH Side** Wedge brake assemblies	Inspect / clean	visually	running freely / no layer of impurities / no excess ware / spring is properly tensioned / Gap between roller & linear is OK	○	
9	Wheel Base Limit switches	Inspect	Visuallly	No loose / actuation & sensing /position is Ok	○	
10	Wheel base Motor	Inspect	Visuallly	Mounting / cable termination OK, Coupling ok / No abnormal noise during running	○	
11	**The cables & termination of all field devices (valve / motors / encoders/ photo sensor , indicators Etc.)**	**Inspect**	**Visualll y**	**No loose / hanging / Twist / Pull**	○	

8.1 Update Document

EN NO : W110-EN-A003-00018

Mahindra Rise

Engineering Notice Release: W110-EN-A003-000189
AD Product Development
Mahindra & Mahindra LTD(A.S)

Created Date 24-Jun-2019
Initiated Date 15-Jul-2019

Engineering Notice No : W110-EN-A003-000189

Released Date : 17-Jul-2019 16:38

Reference Change Req No : DCR044014

Location : AD-NASIK

Implementation Location : M&M-Nasik

Engineering Notice Description : ADDITION OF WHEEL BEARING COVER ON FRONT AXLE KNUCKLE TO PROTECT DUST ENTRY INSIDE WHEEL BEARING

Reason : TO ADDRESS FIELD CONCERN OF WHEEL BEARING FAILURE DUE TO DUST ENTRY INSIDE BEARING

Quantity : -(Regular)

Reason Code : Quality Improvement

Priority : Urgent

Note : TO BE IMPLEMENTED AT SUPPLIER END ON FRONT AXLE ASSY

Affected Models : SCORPIO W110 2WD

Is EN Re Released? :

Parent EN NO :

Life Cycle State : Released

Owner Name : Joshi Girish N (310471)

Engineer : Joshi Girish N (310471)

Approved By : 211714-PATHAK YOGESH

Authorized By : HEAD-PVT-AD_202475-RAUT AVESH

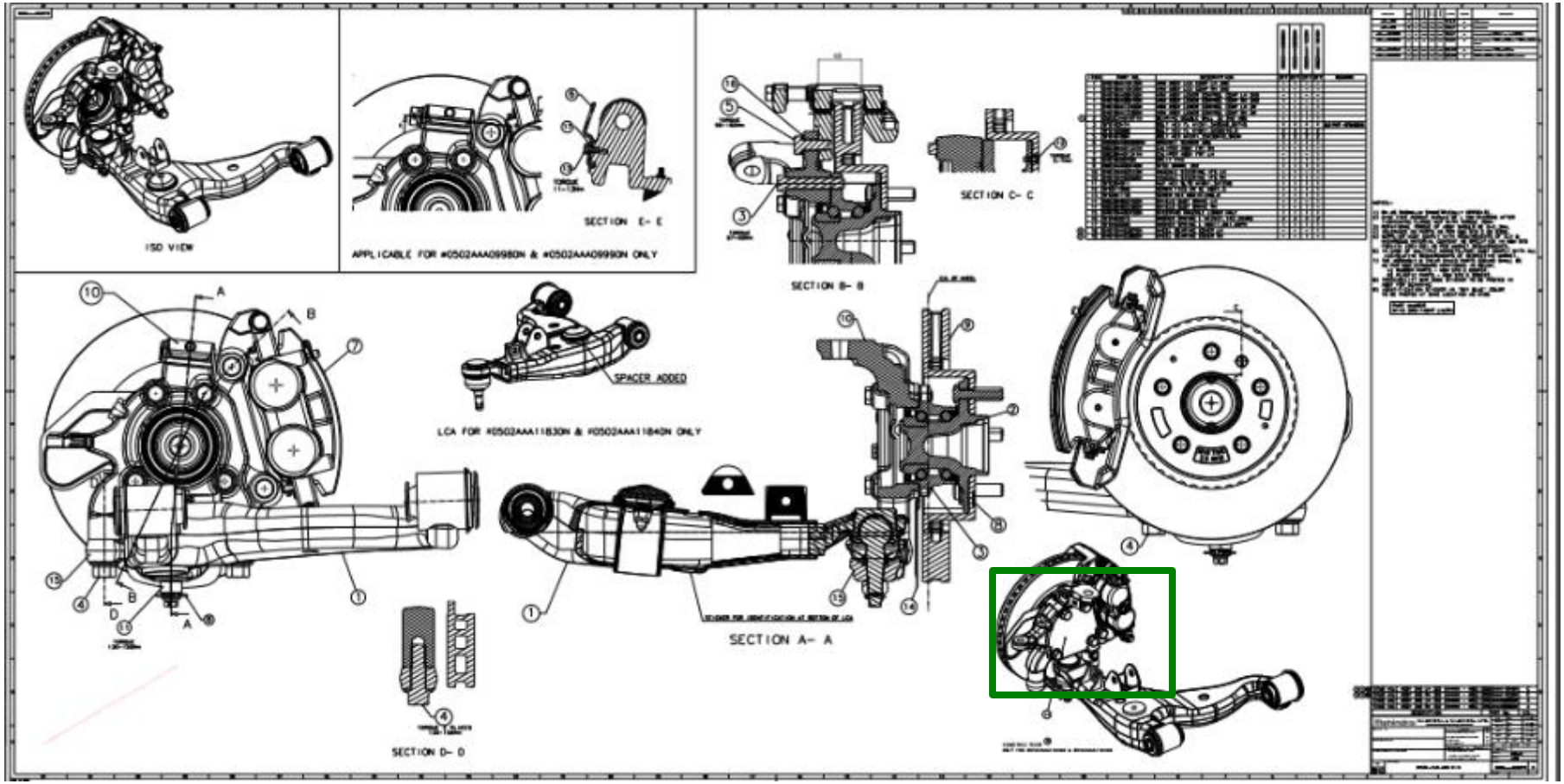
Legends

Add New : A Part/Assy that is getting released through ECN for the very first time

Modified : A Part/Assy that is already released through an ECN and has undergone some changes

8.1 Update Document

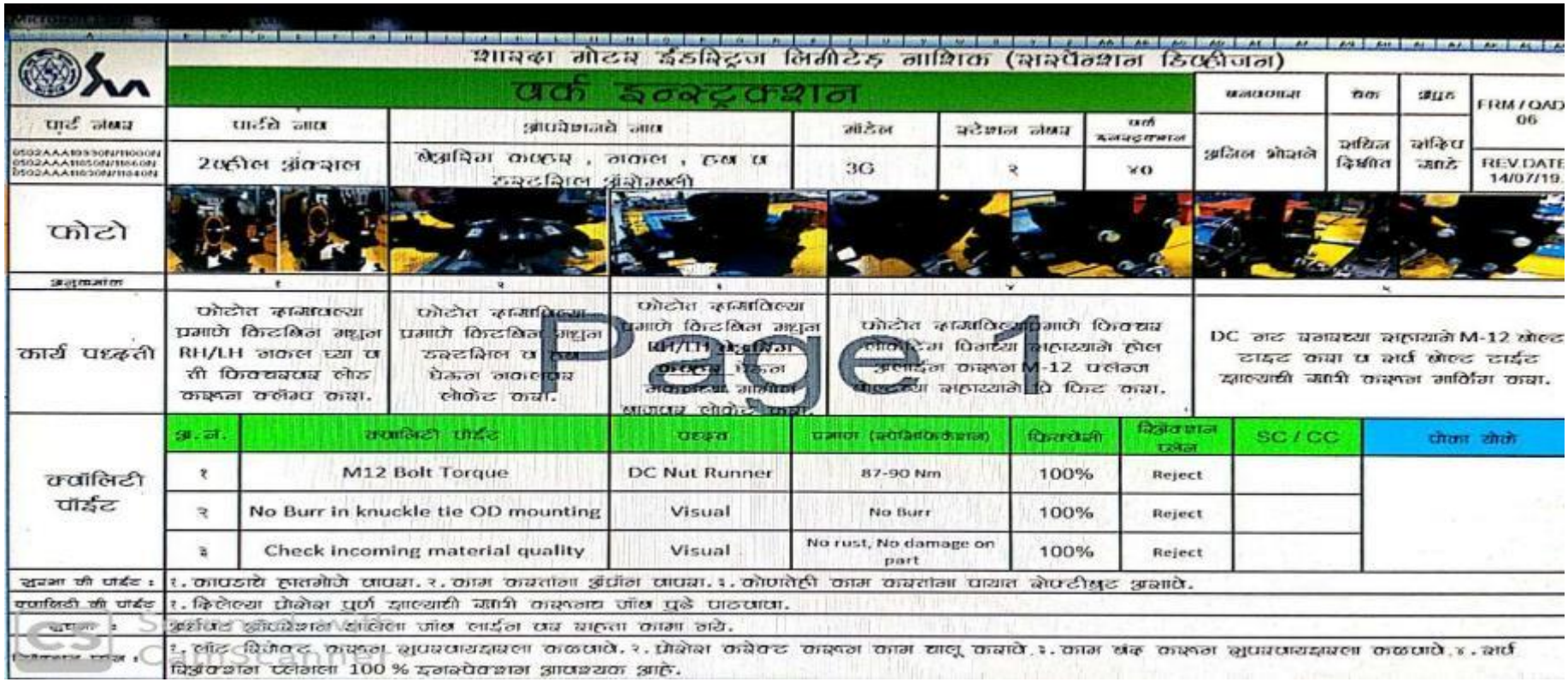
UPDATED DRAWING



8.2 Train people to follow new standards

8.2 Training & Awareness to operators & Inspectors (Supplier) -

Sr.No	Action Decided	Resp.	Actions taken	Status
1	SOP updated at supplier for no burr	DARSHAK	Training & Awareness given to supplier stage operators SOP updated.	**Closed**



शारदा मोटर इंडस्ट्रीज लिमीटेड नाशिक (सस्पेंशन डिव्हीजन)

वर्क इन्स्ट्रक्शन

पार्ट नंबर	पार्टचे नाव	ऑपरेशनचे नाव	मॉडेल	स्टेशन नंबर	वर्क इन्स्ट्रक्शन	बनवणारा	चेक	अप्रुव्ह	FRM / QAD 06
05G2AAA8930N/1160N 05G2AAA8850N/1166N 05G2AAA8830N/1164N	2व्हील अंकशाल	बेअरिंग कव्हर , नकल , हब व इन्स्टलिंग अंकोलवी	3G	२	४०	अमित भोसले	सचिन दिक्षित	संदिप जाधे	REV.DATE 14/07/19.

फोटो

अनुक्रमांक	१	२	३	४	५
कार्य पद्धती	फोटोत दाखविल्या प्रमाणे किटमिन मधून RH/LH नकल घ्या व ती फिक्चरवर लोड करून क्लॅम्प करा.	फोटोत दाखविल्या प्रमाणे किटमिन मधून इन्स्टलिंग व हब घेऊन नकलवर लोकेट करा.	फोटोत दाखविल्या प्रमाणे किटमिन मधून RH/LH [illegible] घेऊन [illegible] लोकेट करा.	फोटोत दाखविल्याप्रमाणे फिक्चर लोकेटिंग पिनाच्या सहाय्याने होल अलाईन करून M-12 प्लेनच [illegible] सहाय्याने टाईट करा.	DC गन बनावच्या सहाय्याने M-12 बोल्ट टाइट करा व शर्ट बोल्ट टाईट झाल्याची खाची करून मार्किंग करा.

क्वॉलिटी पॉईंट

अ.नं.	क्वालिटी पॉईंट	पद्धत	प्रमाण (स्पेसिफिकेशन)	फिक्वेन्सी	रिजेक्शन प्लॅन	SC / CC	फोटो नोंदी
१	M12 Bolt Torque	DC Nut Runner	87-90 Nm	100%	Reject		
२	No Burr in knuckle tie OD mounting	Visual	No Burr	100%	Reject		
३	Check incoming material quality	Visual	No rust, No damage on part	100%	Reject		

सुरक्षा की पॉईंट : १. कापडाचे हातमोजे वापरा. २. काम करताना अंगठी वापरा. ३. कोणतेही काम करताना चश्मा सेफ्टीबूट असावे.

क्वालिटी की पॉईंट : १. दिलेल्या फोटोत पूर्ण झाल्याची खाची करूनच पॉस पुढे पाठवावा.

[illegible] : अस्थित ऑपरेशन झालेला पॉस लाईन वर राहता कामा नये.

[illegible] : १. लोट रिजेक्ट करून सुपरवायझरला कळवावे. २. फोटोत दाखविल्याप्रमाणे काम चालू करावे. ३. काम बंद करून सुपरवायझरला कळवावे. ४. शर्ट रिजेक्शन प्लॅनला 100 % इन्सपेक्शन आवश्यक आहे.

Page 1
CamScanner

9. Summarize and Future Actions

1. Document Learning
2. Horizontal deployment
3. Future action

9. Summarize and Future Actions

9.1 Lessons Learnt :

1. Small small observations like burr (foreign particles) on toe ring are contributing toward major repeat defect, So de not tolerate such failure modes
2. By considering all X factors affecting roller fail in wss ,exact root cause is established.
3. For any new product introduction study past trend history.
4. Involve supplier competency to get new way for analysis
5. DOE tools are very easy tool to identify the root cause & find out solution in a effective manner.
6. Take inputs from similar assembly process ongoing at some other location

9.2 Horizontal Deployment :

1. Horizontal deployment in XUV300

9.3 Future Actions :

1. Input given to process and assembly and supplier teams for future projects.

Thanks !!!!!

- **Team members**
- **Mentors** : Mr. Rahul Narkar
- **MGB coordinator of commodity**:
- **MGB Coordinators**:
- **HR for providing learning opportunity**.
- **MQS team**
- **MQA team**